Michael Remington Coulter

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PROFESSIONAL EXPERIENCE

Arm | Austin, TX

Senior Software Engineer
Apr 2023 - Present

Interconnect Architect | CMN GUI Lead | Tools for internal engineers and external customers.

Recognition: Most Innovative Award, Engineering AI Showcase (Dec 2025), for a project-agnostic multi-agent orchestration and context-gathering system.

- Created all coherent-mesh (CMN) GUI tooling from CMN-800 onward, including Multi-CHI port devices, MCS group configuration, custom routing and mesh deadlock detection interfaces.
- Built a Python NIC-400 XML/CCXML migration engine that reconstructs topology and routes while preserving placement; historical validation recorded 4,130/4,130 valid configurations passing design-rule checks and 5/5 curated invalid configurations rejected.
- Created the NCI/NoC logical canvas and console for configuration logs and design-rule results; built canvas exports and configuration-report scripts.
- Built an Ordered Write Observation wizard for topology, credit/buffer limits and address constraints, with review/apply/restore operations and unit/Playwright tests.
- Implemented grid-aligned physical auto-placement with collision avoidance, grouping and die sizing; tested overlap, containment and adjacency across both renderers.
- Introduced Pixi.js to improve NoC canvas performance and created Playwright test infrastructure for newer non-coherent interconnect tooling.

Software Engineer
Apr 2020 - Apr 2023

- Designed safety-mechanism RTL for lightweight non-coherent interconnects; used waveform viewers to debug testbenches and RTL/IP during RTL and VIP simulation rotations.
- Implemented Flask services to start and stop Jupyter notebooks and persist files in AWS EFS for Pelion's embedded-ML platform; defined PostgreSQL schemas and built end-to-end tests.

Graduate Software Engineer
Jan 2019 - Apr 2020

- Built Verilog FPGA modules and Mbed OS firmware for I2C, GPIO, ADC, DAC and PWM; contributed to shield/form-factor PCBs. 2020 Arm Quality Award (team).
- Built Node.js/Express REST APIs, PostgreSQL schemas and React dashboards to monitor occupancy from thermal cameras and PIR sensors.

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, Verilog, SQL
Systems: Interconnect topology, AMBA, routing, memory maps, design-rule checks, FPGA, Mbed OS
Tools: Angular, Electron, SVG/Pixi, Playwright, Jenkins, Git, LSF, RHEL 8; UVM (earlier work)

EDUCATION

The University of Texas at Austin
Bachelor of Science, Electrical and Computer Engineering | Dec 2018
Primary focus: Computer Architecture and Embedded Systems | Secondary: Integrated Circuits
1st Place (Industry), ECE Capstone Design Award - IoT Wireless Home Security System (2018)